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The Future of Compute in America

Note: For brevity, and due to time constraints, this document is written in point form. It was drafted based on input from (1) AI engineering and security researchers at America's top AI labs, (2) state-level cabinet officials who have to wrestle with burdensome regulations every day to build hyperscaler AI and other national security infrastructure, (3) national security experts who counter some of our adversaries' most advanced operations, and (4) some of the largest private AI data center companies in the United States.

Desired end state

1. Ramp up AI training and inference cluster development in America.

2. Secure AI training and inferencing clusters involved in developing AI systems with probable WMD-like or WMD-enabling capabilities against nation-state espionage and denial/disruption operations.

Context

- This document outlines measures that could kick off a robust Compute in America policy effort that will see us onshore our national security technology, retain and build on U.S. competitive advantage in AI, and accelerate innovation across our industrial base. We should expect to update our approach frequently to keep up with new developments; the field moves too fast for a static framework.

Definitions

- **Advanced AI clusters** are AI-specialized computing clusters that may be geographically distributed (but still with dedicated point-to-point connections; no need to include all data centers connected over standard internet fiber connections) either:
  - With aggregate power supply of >50MW or;
  - Expected to house AI processors in the aggregate capable of >2x10^20 FLOP/s with sparsity, at any numerical representation or;
  - That have trained, deployed, or stored models shown to be capable of independently producing strategic classified U.S. secrets when prompted, as determined by a relevant U.S. Government entity.

- **Critical AI clusters** are AI-specialized computing clusters that may be geographically distributed (but still with dedicated point-to-point connections; no need to include all data centers connected over standard internet fiber connections) either:
  - With aggregate power supply of >200MW or;
  - Expected to house AI processors in the aggregate capable of >8x10^20 FLOP/s with sparsity, at any numerical representation or;
  - That have trained or deployed models shown to be capable of WMD development (either autonomously or by supporting non-experts with such development), or shown capable of independently producing strategic classified U.S. secrets deemed by a relevant U.S. Government entity to be of critical importance to U.S. national security.

Government support for American infrastructure

Deregulation and tax incentives

- Some types of security research – such as research focused on developing flexHEG technologies, or nation-state-level red teaming and hardening of frontier AI infrastructure – are badly needed to ensure the security of domestic frontier AI development, and hard to fund commercially because security is viewed as a cost center rather than a competitive advantage at leading labs. The U.S.

Government could subsidize this work via grants and tax incentives. Investments frontier labs make in hardening against nation state espionage, theft, and sabotage are investments in U.S. national security.

- The government could invoke DPA Title III to provide NEPA exemptions to advanced and critical AI clusters and related dedicated energy infrastructure that meet requirements that will be defined in the Government Support for Data Center Security section.

- The government could introduce categorical exclusions to NEPA for no-impact preliminary activities on these projects, like design, drilling boreholes, collecting soil samples. This would help unlock early DOE Loan Programs Office financing.

Energy infrastructure & permitting

- The President could declare AI-related power infrastructure as critical to national security, enabling expedited processes.

- The President could direct federal agencies to establish a 3-month maximum timeline for power infrastructure project approvals ("idea-to-construction").

- The President could direct agencies to revise the definition of energy "emergency" to include supporting rapid development of infrastructure in a competitive geopolitical context.
    - "Energy emergency" provisions are the workaround to environmental opposition to carbon-intensive energy sources. In litigation against a power generation build, emergency provisions (essential service provisions, greater good, etc.) can be called on by the courts to enable energy generation.

- The government could mandate consolidated/unified permitting processes across federal agencies to eliminate redundant reviews. Refined processes should be directly informed by a working group with representation by industry and the states to ensure the points of greatest friction are removed.

Environmental review reform

- The President could order agencies to reform environmental review processes to enable rapid approvals, and establish expedited environmental review processes for AI/compute-related power projects.

- The President could direct creation of limits on litigation options for designated critical infrastructure projects.
    - In particular, expedited court decisions. For example: a mandate that cases need to be concluded within 30 days. Every mechanism for delay should be time-bounded.

Workforce development

- The Department of Labor could establish accelerated training programs for utility/power infrastructure trades.

- The Department of Labor could order creation of a national apprenticeship program focused on power infrastructure construction and sponsor state-level programs (e.g. grants) calibrated for the most relevant trades and energy industries in their territory. Federal apprenticeship programs could include a one-year fellowship with a key department or agency to ensure the executive branch remains attuned to the industry's needs.

Critical infrastructure security

- The President could order enhanced physical and cybersecurity requirements for power infrastructure serving AI compute facilities.

- The Department of Homeland Security could direct coordination between federal agencies on infrastructure security evaluations.

- The Department of Homeland Security could mandate the creation of information-sharing protocols between public/private entities on infrastructure security.

- DHS and DOE, with the support of the CIA and NSA could study the feasibility of securing fiber communications between data centers to form geographically distributed clusters. This is very time-sensitive, since it affects data center design decisions that are difficult to reverse once made.

    - This study should consider the trade-offs between (1) the risks introduced by dark fiber security vulnerabilities and (2) the access to power that geographically distributed clusters (featuring sites linked together by dark fiber) would provide.

Resource access

- The President could direct agencies to expedite access to critical minerals needed for infrastructure development, granting priority access to critical and advanced cluster component supply chains.

- Same dereg/permitting approvals terms as above (<3 months, etc.) could apply here to critical minerals.

- In addition, the President could suspend or mitigate the FCPA abroad, which is increasing the effective cost of U.S.-based companies to explore and extract critical minerals outside the United States.
    - Note that FCPA is Congressionally mandated so recommend a WH legal review to understand to what extent this can in fact be mandated.

- The President could order a review/reform of restrictions on domestic mining and resource extraction.

- The President could order a review of legal recourse/penalties for weaponization of lawfare with increased penalties for conspiring with or involvement with foreign entities. In addition to increased penalties, legal costs could be borne by the entity under review.

Government support for secure AI data centers in America

Personnel security

- The President could direct the U.S. Government to develop (90 days) and offer (120 days) a timely and effective background check option for sensitive roles at advanced AI clusters, critical AI clusters, and AI labs, or develop the criteria that these entities should/must use to perform such background checks and make hiring decisions based on them.

Government support for data center security

- To maintain our lead in AI, we must build the AI data centers of the future in America. To avoid our adversaries simply stealing our advances, those data centers need to be secure against adversary infiltration. But data centers take a long time to build and cannot be retrofitted for adequate security at this level after the fact. This means we need to build this infrastructure securely now.

- There could be some “best effort” U.S. Government-incentivized (tax incentives, funding, resources made available, etc.) directed at strengthening the security of existing clusters. Otherwise America’s future AI models — that will be trained on today's clusters — will just end up in the CCP’s hands automatically.

- The President could instruct CISA and the AISI, with technical support from NSA, CIA, and CYBERCOM, to define security requirements for inference clusters, since these will also hold sensitive model weights while being potentially much smaller than training clusters.

- The President could appoint an “AI infrastructure czar” to coordinate policy to accelerate the build-out of secure data centers. The steps described here will require the use of a broad set of federal authorities across different agencies, as well as coordination with private companies.

- The President could direct Commerce, State, and DOD to investigate the prospect of setting up a version of the DOD’s Trusted Foundry Program for advanced AI chips fabbed on leading nodes, with its mandate to include working with international foundries like TSMC who have domestic footprints. The idea would be to later be able to offer government subsidies/incentives for using hardware fabbed in trusted AI foundries.

- The DOE could establish a working group to identify points of contact on data center security who can advise the designers, builders, and operators of advanced AI clusters and critical AI clusters on vulnerabilities to nation-state exfiltration, sabotage, and denial ops. This working group could operate under the understanding that the United States may ultimately require that training runs and deployments above a certain threshold be carried out within security-compliant clusters. This working group could cover, for example:
    - Cyber, physical, RF/electronic warfare and personnel security.

    - Passive data collection techniques.
    - TEMPEST standards (may include sharing classified standards in whole or in part, if deemed necessary).

- The President could instruct the relevant government entities, with technical support from NSA, CIA, CYBERCOM to take an active role in advanced AI data center and critical AI data center security:
    - By developing best practices and sharing tactics, techniques, and procedures to fortify advanced AI and critical AI clusters with select private sector partners.
    - By developing counterintelligence playbooks to address risk of nation state exfiltration and denial attacks against clusters and labs based on such practices as passive data collection and active monitoring of adversary efforts.
    - By coordinating these efforts closely with (classified) denial operations.

- The President could instruct the NSA, CIA, and CYBERCOM to conduct pen-testing of current and future advanced AI clusters to map vulnerabilities to physical/cyber/personnel access points. The results of these exercises should be shared with labs/data centers if deemed supportive.

Government support for AI hardware security

- The President could instruct the NSA, CIA, CYBERCOM to identify points of contact on chip security (and hardware more broadly) who can advise on design for next-generation chips with a view to securing critical American national security IP. These agencies could create a working group to coordinate with chip design firms and fabs on chip security.

- The President could instruct DOE National Laboratories, DARPA, IARPA to launch new research programs focused on radically accelerating security for AI hardware devices, including cluster-scale confidential computing, and making advanced AI hardware devices highly resilient to both invasive and non-invasive physical attacks from sophisticated actors (including possible tampering at every stage in the supply chain). This should be done in collaboration with the relevant private sector entities.

- The President could instruct NIST to develop new, prescriptive security standards specifically for AI hardware devices, working in collaboration with industry partners and agencies including NSA, CIA, and CYBERCOM. Unlike current standards such as FIPS 140-3 which provide general security objectives, these new standards should detail specific technical requirements, implementation measures, and verification methods required to protect AI hardware against sophisticated attackers.

Indications and warnings

Situational awareness for cloud computing

- To learn who is using certain compute resources and for what purpose, the President could have Commerce/BIS:
    - Establish an advanced AI chips registry to ensure chips are delivered to intended destinations and remain subject to monitoring. Combine with either random inspections or location verification features to make this actually useful for enforcement.
    - Implement reporting and evaluation obligations for AI clusters used to train models above a training compute threshold (10^26 FLOP), or with total power consumption over the “advanced AI cluster” threshold of 50MW. (For context: the capex associated with a 50MW cluster is over a billion dollars.)
    - Establish a know-your-customer (KYC) identification and screening for access to high-volume compute usage to verify the identity and legitimacy of users.
    - Require commitments from recipient countries and entities, such as restrictions on sharing model weights, access to U.S. natsec red teamers, other security standards.
        - The Chinese are doing deals with offshore non-U.S. data centers for AI training cloud services, and essentially funding these facilities (e.g. Bytedance in Malaysia).
    - Require regular reporting and auditing of compute usage to identify patterns of misuse and inform future policy decisions.
    - Establish clear consequences for violating these commitments and restrictions, such as the revocation of access to AI compute resources, sanctions, and more.

Identifying future requirements

- The President could order the formation of a task force jointly with industry stakeholders to look into and produce ongoing/evolving recommendations/requirements for:
    - Infrastructure for ultra-secure advanced AI and critical AI clusters.
    - Defining public and classified standards for data center security.
    - Identifying security considerations associated with the development of technologies allowing for distributed training of critical AI models in highly decentralized settings (e.g. DiLoCo, Intellect-1, Nous Research-type schemes).

Data center security requirements

- The Secretary of Commerce, in coordination with the Director of National Intelligence and the Secretary of Defense, could establish a task force within 30 days to develop comprehensive data center security standards for facilities housing advanced AI clusters (>50MW) and critical AI clusters (>200MW).
    - This task force could be required to develop distinct security frameworks for advanced AI and critical AI clusters, with critical AI facilities requiring security measures at least as rigorous as those required for Top Secret/Sensitive Compartmented Information facilities, addressing physical security, cybersecurity, supply chain integrity, and access controls.

- The U.S. Government could require companies building data center shells meeting the frontier AI power threshold (>200MW) to obtain BIS approval before breaking ground on a new build if they have received investment.